JouleBridge Runtime Architecture

JouleBridge Papers

ABSTRACT

JouleBridge is the runtime, proof, and coordination layer for distributed energy sites. The system is designed so control decisions stay close to the assets while the resulting actions remain reviewable later by another team.

This paper explains the runtime architecture as an operating system for real sites: how the Asset Agent, Site Kernel, Portfolio Orchestrator, and Console divide responsibilities without collapsing local execution into a cloud-only product story.

1. SYSTEM OBJECTIVE

The architecture is built around one durable question: how should a mixed-vendor site coordinate chargers, batteries, solar, flexible loads, and meters while still leaving behind a record that others can trust later?

That question pushes the design toward local control, explicit trust boundaries, and proof attached to the operating path rather than reporting layered on after the fact.

2. RUNTIME SPLIT

The Asset Agent sits close to the equipment and adapter layer. The Site Kernel converts policy, limits, and site state into feasible local actions. The Portfolio Orchestrator distributes bundles, receives proofs, and gives the fleet view. The Console presents the operator surface.

Each runtime has a narrow job. None of them should pretend to be the whole platform on its own.

3. LOCAL-FIRST COORDINATION

JouleBridge keeps the coordination loop at the site because cloud latency, tunnel fragility, and vendor heterogeneity all get worse when local autonomy is removed. The system should keep dispatch running even when upstream connectivity is degraded.

Local execution is therefore a product requirement, not an optimization detail.

4. ADAPTER AND POLICY BOUNDARIES

Adapters translate mixed-vendor equipment into a stable internal shape. Policy then evaluates whether the requested action is feasible, safe, and worth routing. This boundary matters because weak adapter behavior or vague policy handling quickly contaminates later operations.

A site runtime only becomes trustworthy when ingress and policy decisions are legible.

5. PROOF CHAIN

JouleBridge attaches proof to the operating path. Intent, dispatch, acknowledgement, measurement, and outcome are shaped into a record that can be verified later without informal reconstruction.

The goal is practical defensibility. Another team should be able to answer what the site believed, what it did, and what happened next.

6. DEPLOYMENT POSTURE

The preferred deployment model is partner hardware first. The commercial target is a repeatable install path on real site infrastructure before broader platform claims are made.

That is why DepotOS starts with EV fleet depots rather than trying to tell every distributed-energy story at once.

CONCLUSION

JouleBridge should be read as a site runtime with a proof chain, not as a dashboard wrapper. Stronger local coordination and a cleaner operational record make the later cloud view more credible and easier to operate.